Test 2: Wed, Nov 5, in class.

Fourier transform methods

- Given a smooth function $f : \mathbb{R} \to \mathbb{R}$ that decays rapidly (faster than $\frac{1}{(1+|x|)^p}$ for any $p > 0$.), define the Fourier transform of f,

$$\mathcal{F}[f] = \hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} \, dx$$

Inversion formula:

$$\mathcal{F}^{-1}[\hat{f}] = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} \, d\omega$$

$e^{i\omega x} = \cos(\omega x) + i \sin(\omega x)$

(Fourier series:

$$f(x) = \sum_{n=-\infty}^{\infty} a_n e^{-i\omega_n x}$$

$f : [0, L] \to \mathbb{R}$)

If $f : \mathbb{R}^n \to \mathbb{R}$, $\hat{f} : \mathbb{R}^n \to \mathbb{C}$

$$\hat{f}(\omega) = \int_{\mathbb{R}^n} f(x) e^{-i\omega \cdot x} \, dx$$

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\omega) e^{i\omega \cdot x} \, d\omega$$

Derivatives: $n = 1$

$$\widehat{f'} = \int_{-\infty}^{\infty} f'(x) e^{-i\omega x} \, dx$$

$$= -\int_{-\infty}^{\infty} f(x) e^{-i\omega x} (-i\omega) \, dx + \cancelto{0}{f(x) e^{-i\omega x} \Big|_{-\infty}^{\infty}}$$

$$= i\omega \int_{-\infty}^{\infty} f(x) e^{-i\omega x} \, dx = i\omega \hat{f}(\omega)$$

$n \geq 2$ $\quad \widehat{\partial_{x_j} f} = i\omega_j \hat{f}(\omega)$

$$\widehat{\partial_{x_j}\partial_{x_j} f} = (i\omega_j)^2 \hat{f}(\omega) = -\omega_j^2 \hat{f}(\omega)$$

<u>Convolution:</u>

Recall: $\quad (f * g)(x) = \int_{\mathbb{R}^n} f(x-y) g(y)\, dy$

$$\widehat{f * g} = \hat{f}\hat{g} \longrightarrow (f*g)(x) = \mathcal{F}^{-1}\left[\hat{f}\hat{g}\right]$$

Poisson's Equation:

$$-\Delta u = f \qquad x \in \mathbb{R}^n.$$

Take $\mathcal{F}$ of the equation:

$$\mathcal{F}[-\Delta u] = \mathcal{F}\left[-\sum_{j=1}^{n} \partial_{x_j}^2 u\right] = -\sum_{j=1}^{n} \mathcal{F}[\partial_{x_j}^2 u] = -\sum_{j=1}^{n} (i\omega_j)^2 \hat{u}$$

using linearity of $\mathcal{F}$

$$= \sum_{j=1}^{n} \omega_j^2 \hat{u}(\omega)$$

$$= |\omega|^2 \hat{u}(\omega)$$

$$|\omega|^2 \hat{u} = \hat{f}$$

Algebraic equation between $\hat{u}$ & $\hat{f}$. $\rightarrow$ $\hat{u}(\omega) = \frac{1}{|\omega|^2} \hat{f}(\omega)$

Take $\mathcal{F}^{-1}$:

$$\mathcal{F}^{-1}[\hat{u}] = \mathcal{F}^{-1}\left[\frac{1}{|\omega|^2}\hat{f}\right]$$

$$u = \mathcal{F}^{-1}\left[\frac{1}{|\omega|^2}\right] * \mathcal{F}^{-1}[\hat{f}]$$

by Convolution Rule

$$\mathcal{F}^{-1}\left[\frac{1}{|\omega|^2} + \hat{f}\right]$$
$$= \mathcal{F}^{-1}\left[\frac{1}{|\omega|^2}\right] + \mathcal{F}^{-1}[\hat{f}]$$

$$u(x) = \mathcal{F}^{-1}\left[\frac{1}{|\omega|^2}\right] * f$$

HW: $\mathcal{F}^{-1}\left[\frac{1}{|\omega|^2}\right] = c|x|^{2-n}$, $n \geq 3$

$$\Rightarrow u(x) = c|x|^{2-n} * f(x) = c\int_{\mathbb{R}^n} |x-y|^{2-n} f(y)\,dy.$$

1D Heat Equation:

$$\begin{cases} u_t - u_{xx} = 0 \\ u(x,0) = g(x) \end{cases}$$

Assume u, g are smooth, rapidly decaying.

Take $\mathcal{F}$ of the PDE:

$$\mathcal{F}_x[u_t] - \mathcal{F}_x[u_{xx}] = 0$$

$$\mathcal{F}[u_t] = \int_{-\infty}^{\infty} u_t(x,t)\, e^{-i\omega x}\, dx$$

$$= \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u(x,t)\, e^{-i\omega x}\, dx$$

$$= \hat{u}_t(\omega, t)$$

$$\mathcal{F}[u_{xx}] = (i\omega)^2 \hat{u} = -\omega^2 \hat{u}(\omega, t)$$

$$\hat{u}_t + \omega^2 \hat{u} = 0.$$

For every $\omega \in \mathbb{R}$, this is an ODE in t for $\hat{u}(\omega, t)$

$$\hat{u}(\omega, 0) = \mathcal{F}[u(x,0)] = \hat{g}(\omega)$$

$$\begin{cases} \hat{u}_t = -\omega^2 \hat{u} \\ \hat{u}(\omega,0) = \hat{g}(\omega) \end{cases}$$

Linear ODE:

$$\hat{u}(\omega,t) = \hat{u}(\omega,0)\, e^{-\omega^2 t}$$

$$= \hat{g}(\omega)\, e^{-\omega^2 t}$$

$$u(x,t) = \mathcal{F}_x^{-1}\left[\hat{g}(\omega)\, e^{-\omega^2 t}\right]$$

$$= g(x) * \mathcal{F}^{-1}\left[e^{-\omega^2 t}\right]$$

$$\mathcal{F}^{-1}\left[e^{-\omega^2 t}\right]$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\omega^2 t}\, e^{i\omega x}\, d\omega$$

$$e^{-\omega^2 t}\, e^{i\omega x} = e^{-t\left(\omega^2 - \frac{i\omega x}{t}\right)}$$

$$= e^{-t\left(\omega^2 - \frac{i\omega x}{t} + \left(\frac{ix}{2t}\right)^2\right) + t\left(\frac{ix}{2t}\right)^2}$$

$$\mathcal{F}^{-1}\left[e^{-\omega^2 t}\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-t\left(\omega - \frac{ix}{2t}\right)^2}\, e^{-x^2/4t}\, d\omega$$

$$= \frac{1}{2\pi}\, e^{-x^2/4t}\int_{-\infty}^{\infty} e^{-t\left(\omega - \frac{ix}{2t}\right)^2}\, d\omega$$

$$\int_{-\infty}^{\infty} e^{-t\left(\omega - \frac{ix}{2t}\right)^2} d\omega = \int_{-\infty}^{\infty} e^{-z^2} \frac{1}{\sqrt{t}} dz$$

$$\boxed{\int_{-\infty}^{\infty} e^{-z^2} dz = \sqrt{\pi}}$$

$$\boxed{\begin{aligned} z &= \sqrt{t}\left(\omega - \frac{ix}{2t}\right) \\ dz &= \sqrt{t}\, d\omega \end{aligned}}$$

$$= \frac{\sqrt{\pi}}{\sqrt{t}}$$

$$\mathcal{F}^{-1}\left[e^{-\omega^2 t}\right] = \frac{1}{2\pi} e^{-x^2/4t} \frac{\sqrt{\pi}}{\sqrt{t}} = \frac{1}{2\sqrt{\pi t}} e^{-x^2/4t}$$

$$= \frac{1}{\sqrt{4\pi t}} e^{-x^2/4t} = \Phi(x,t)$$

Heat Kernel.

$$\boxed{u(x,t) = g * \Phi = \int_{-\infty}^{\infty} \frac{e^{-|x-y|^2/4t}}{\sqrt{4\pi t}} g(y)\, dy}$$

HW: $u_t - u_{xx} = f$

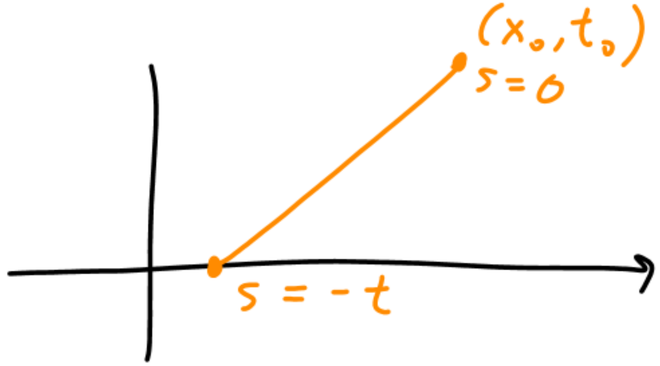
Wave Eqn: $\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x,0) = g \\ u_t(x,0) = h \end{cases}$

$$\begin{cases} \hat{u}_{tt} + \omega^2 \hat{u} = 0 \\ \hat{u}(x,0) = \hat{g} \\ \hat{u}_t(x,0) = \hat{h} \end{cases} \longrightarrow \hat{u}(\omega,t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$\cdots$

$$u_{tt} - u_{xx} = u^2$$

General first-order PDEs



Recall: Linear Transport:

$$a u_t + b u_x + c u = 0$$

a, b, c are numbers:

characteristics $(x,t) = (x_0 + bs, t_0 + as)$

$$z(s) = u(x_0 + bs, t_0 + as)$$

$$z'(s) = b u_x(x_0 + bs, t_0 + as) + a u_t(x_0 + bs, t_0 + as) = -c z(s)$$

$$z' = -cz \longrightarrow z(s) = e^{-cs} z(0) \qquad s = -t_0/a$$

$$u(x_0, t_0) = u\left(x_0 - \frac{b t_0}{a}, 0\right) e^{-c t_0}$$

What if a, b, c are functions of x, t?

$$\begin{cases} \boxed{a(x,t)}\, u_t + \boxed{b(x,t)}\, u_x + c(x,t)\, u = 0 \\ u(x,0) = g(x) \end{cases}$$

characteristics par. by $(x(s), t(s))$

$$z(s) = u(x(s), t(s))$$

$$\frac{d}{ds} z(s) = u_x \boxed{\frac{dx}{ds}} + u_t \boxed{\frac{dt}{ds}} = u_x b + u_t a = -cu$$

$$\boxed{z'(s) = -c(x(s), t(s))\, z(s)}$$

$$\boxed{\begin{aligned} x'(s) &= b(x(s), t(s)) \\ t'(s) &= a(x(s), t(s)) \end{aligned}}$$

Characteristic ODEs

Linear, Variable-Coefficient First-order ODE system.

Example

$$\begin{cases} u_t + 3t^2 u_x = 2t\,u \\ u(x,0) = g(x) \end{cases}$$

$$z(s) = u(x(s), t(s))$$

$$\frac{d}{ds} z(s) = u_x \frac{dx}{ds} + u_t \frac{dt}{ds}$$

$$\frac{dt}{ds} = 1 \rightarrow t = t_0 + s$$

$$\frac{dx}{ds} = 3t^2 = 3(t_0+s)^2$$

Integrate in s:

$$x(s) = (t_0+s)^3 + c$$

$$x_0 = x(0) = (t_0+0)^3 + c$$

$$\rightarrow c = x_0 - t_0^3$$

$$x(s) = (t_0+s)^3 + x_0 - t_0^3 .$$